



REVOFIN
SMART FINANCES

Risk Optimization: The Debt Consolidation Strategy

Analytical Report



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METHODOLOGY

Business Understanding

Understanding business problems, setting objectives, and project scope

Cleaning & Aggregation

Collecting data and ensuring data quality with SQL

Analysis & Visualization

Performing data analysis to generate insights with SQL and Tableau

Insights & Recommendation

Effectively communicating insights and recommendations to solve problems

Disclaimer :

Any reference to its services or data analysis methods is illustrative only. This scenario is designed to showcase the skills of a data analyst and is not associated with any financial institution. Any similarities to actual companies are purely coincidental.



01 Business Understanding

ABOUT US, BUSINESS PROBLEM,
ROOT CAUSE ANALYSIS

ABOUT US



RevoFin is a **Fintech company** that provides innovative lending solutions for **individuals** and **small businesses**. It revolutionizes the borrowing experience using **technology** and **data-driven insights**. Its primary revenue model is based on transaction fees and loan interest.

BUSINESS PROBLEM

How RevoFin can optimize its risk-based pricing strategy for a **\$2.8 billion** portfolio with **176,087 customers**, where analysis shows that debt consolidation customers (58.96% of low performers) have a distinct risk profile that is not yet reflected in pricing differentiation within 6 months.

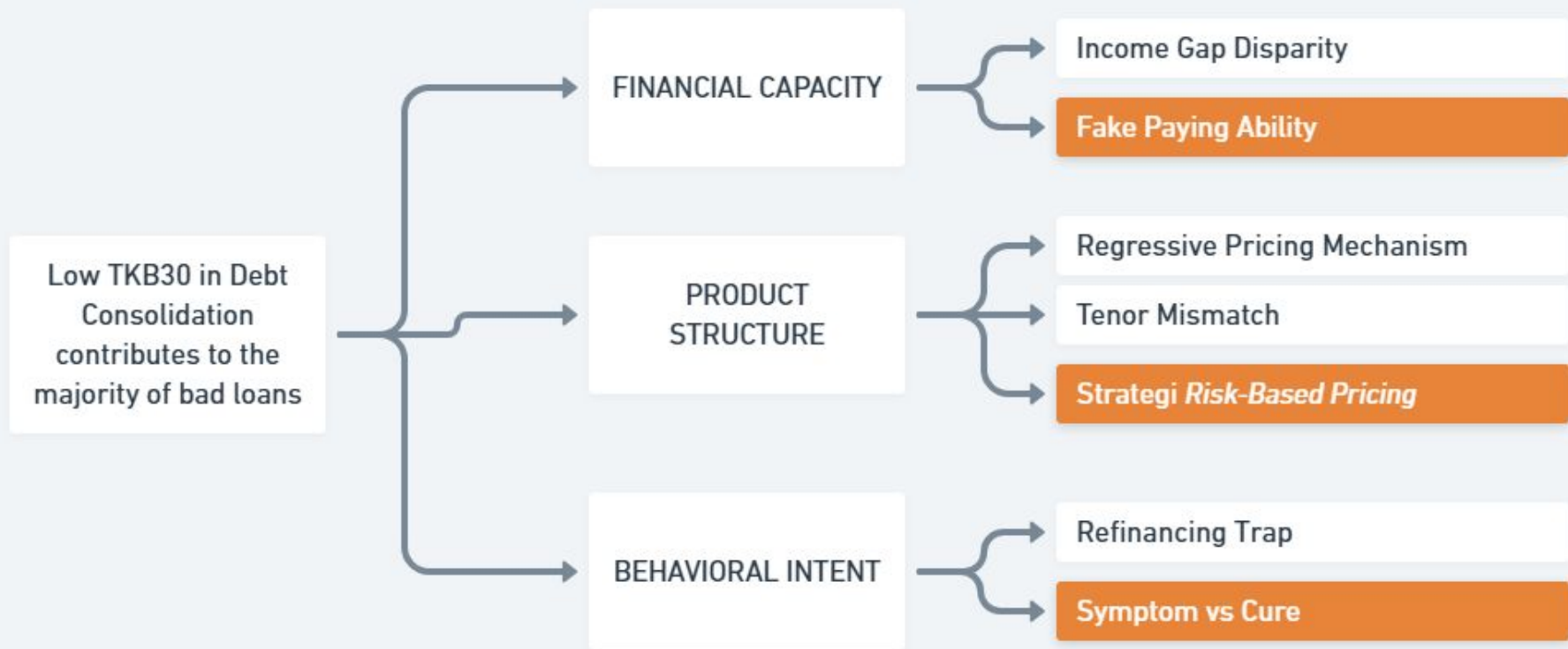
OBJECTIVE

- **Outlining** the **total outstanding loan amounts**,
- Highlighting **key risk metrics**, and
- Examining both **loan characteristics** and **customer** attributes to uncover meaningful patterns or potential risk areas.

TOOLS & METHODS

- Data Processing - SQL (Cleaning Data & Aggregation)
- Visualization - Tableau
- Statistical Analysis - EDA, Cohort, Correlation, Customer Segmentation

Root Cause Analysis





02 Analysis

EDA, INSIGHT, DATA VISUALIZATION

LOAN OVERVIEW PORTFOLIO



176.087

CUSTOMER



\$2.801.078.850

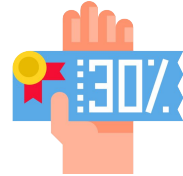
OUTSTANDING



\$2.800.883.575

ENDING NET
RECEIVABLES

12 = \$195.275
Default

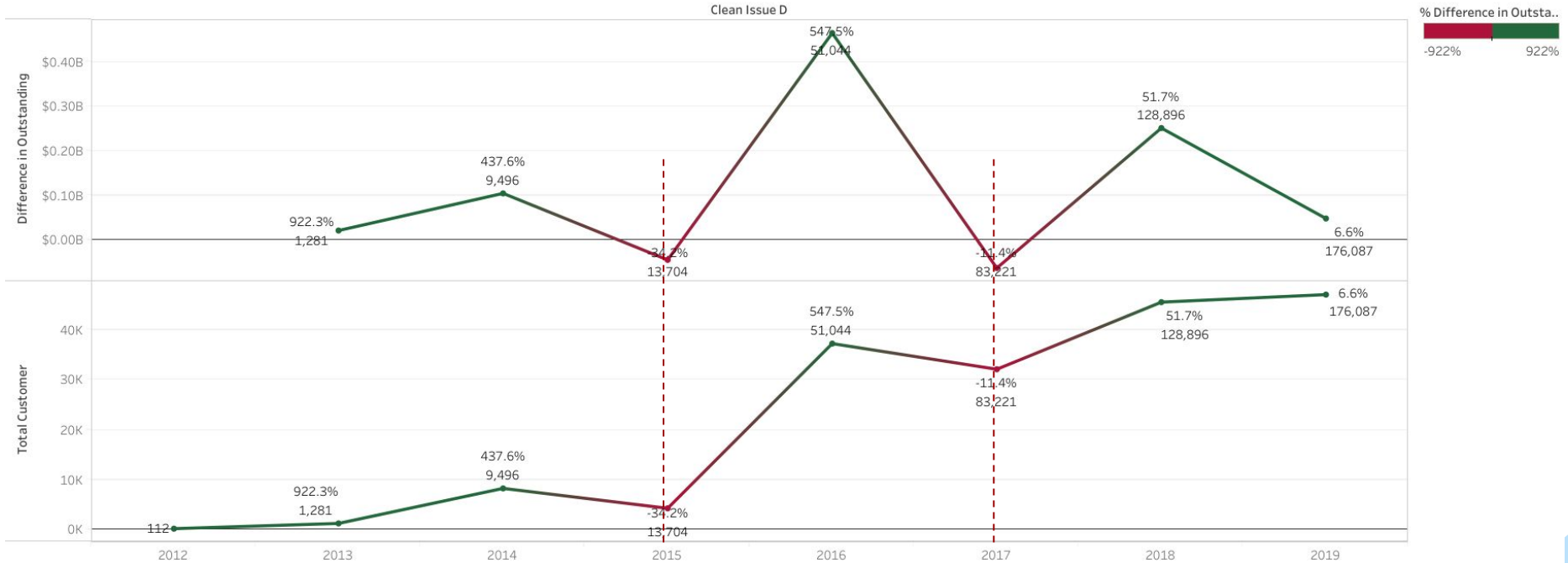


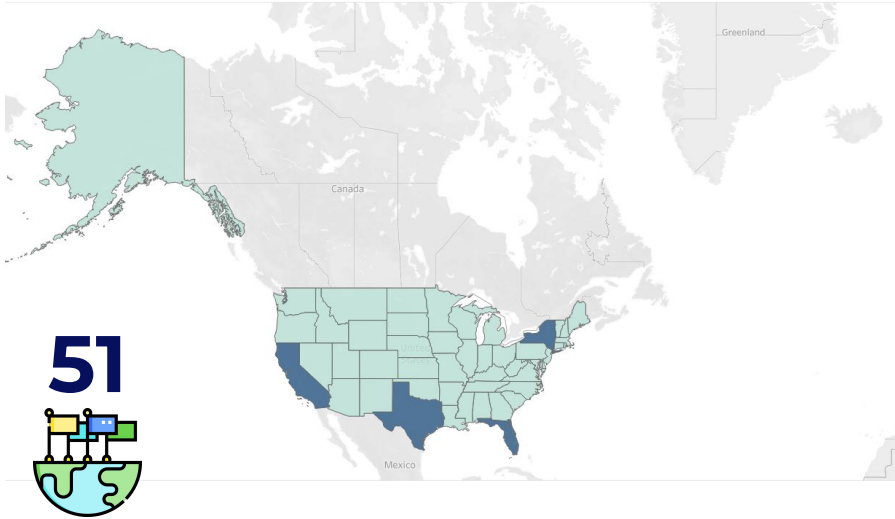
98.11%

TKB 30

Business Growth

Business Growth





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Cohort (Heatmap)

Month of Cl..	2012	2013	2014	2015	2016	2017	2018	2019	TKB30
January		96.84%	94.86%	99.79%	96.56%	97.40%	98.97%	99.47%	93.28% 100.00%
February		97.28%	95.60%	97.48%	96.26%	96.69%	98.46%	99.85%	
March		100.00%	97.95%	98.23%	97.01%	96.75%	98.70%	99.79%	
April		97.09%	93.28%	98.07%	97.42%	96.96%	98.92%	97.90%	
May		94.10%	97.70%	96.75%	97.75%	96.18%	99.54%	98.36%	
June		97.69%	96.43%	95.74%	97.61%	97.21%	99.70%	98.27%	
July		97.07%	96.25%	98.28%	97.08%	96.99%	97.36%	98.32%	
August	100.00%	100.00%	95.76%	96.56%	97.66%	97.10%	98.01%	98.70%	
September	96.69%	95.66%	95.82%	98.41%	97.51%	97.19%	98.26%	98.40%	
October	100.00%	94.55%	97.29%	97.12%	98.76%	97.70%	98.31%	98.81%	
November	94.96%	98.31%	97.45%	98.55%	98.94%	97.66%	98.31%	98.98%	
December	94.74%	96.33%	96.58%	95.39%	99.03%	98.06%	99.39%	99.30%	

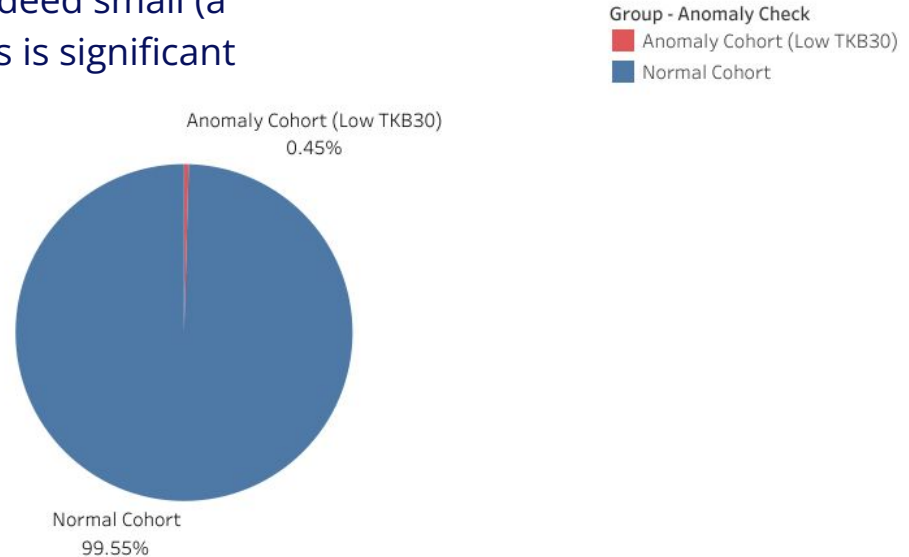
Portfolio Health Overview:
Strong but Needs Attention



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Anomaly vs Normal Customer

In terms of quantity, this Anomaly group is indeed small (a minority), but the financial impact of its losses is significant because its TKB30 has plummeted.

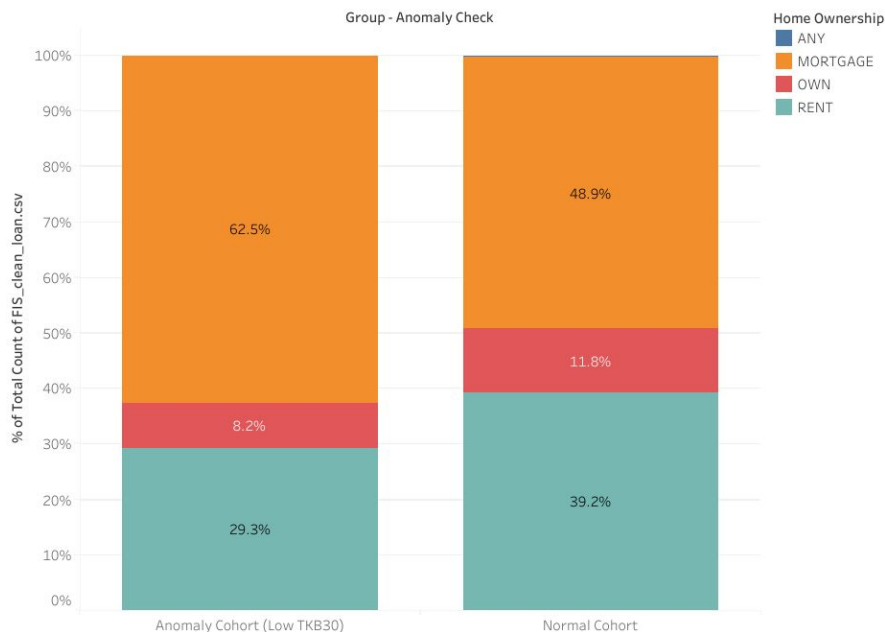


Home Ownership Profile

Anomaly vs Normal Customer



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The Mortgage Trap

these customers take out new loans (debt consolidation), their cash flow immediately becomes **negative**. Their entire **salary** goes toward **mortgage payments** & **new debt payments**.

Renters aren't the villain

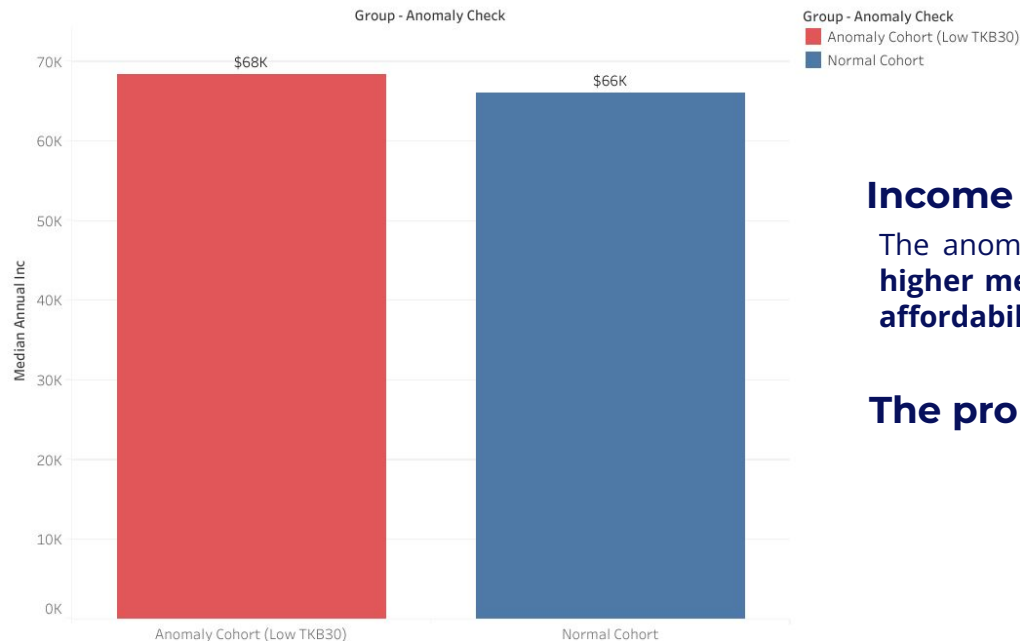
Rejecting a loan application simply because the applicant is "**still renting**" is the **wrong strategy**. The problem is **not the status** of the house, **but the amount of money left**.

The "OWN" is Safety Net

Customers who have paid off their mortgages are the safest. They have more disposable income to pay off their debts to RevoFin.

Income Analyst

Anomaly vs Normal Customer



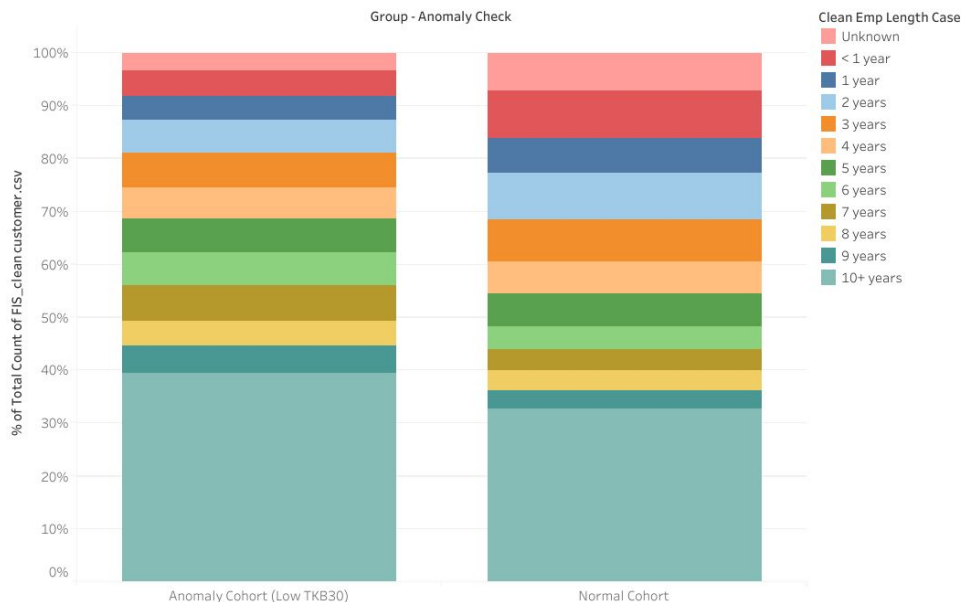
Income is not the main reason for low TKB30

The anomaly cohort (those with **low TKB30**) actually **had a higher median income**, meaning that the problem **was not affordability**.

The problem may lie in non-income factors

Employment Length Analyst

Anomaly vs Normal Customer



Stability does not guarantee ability to pay

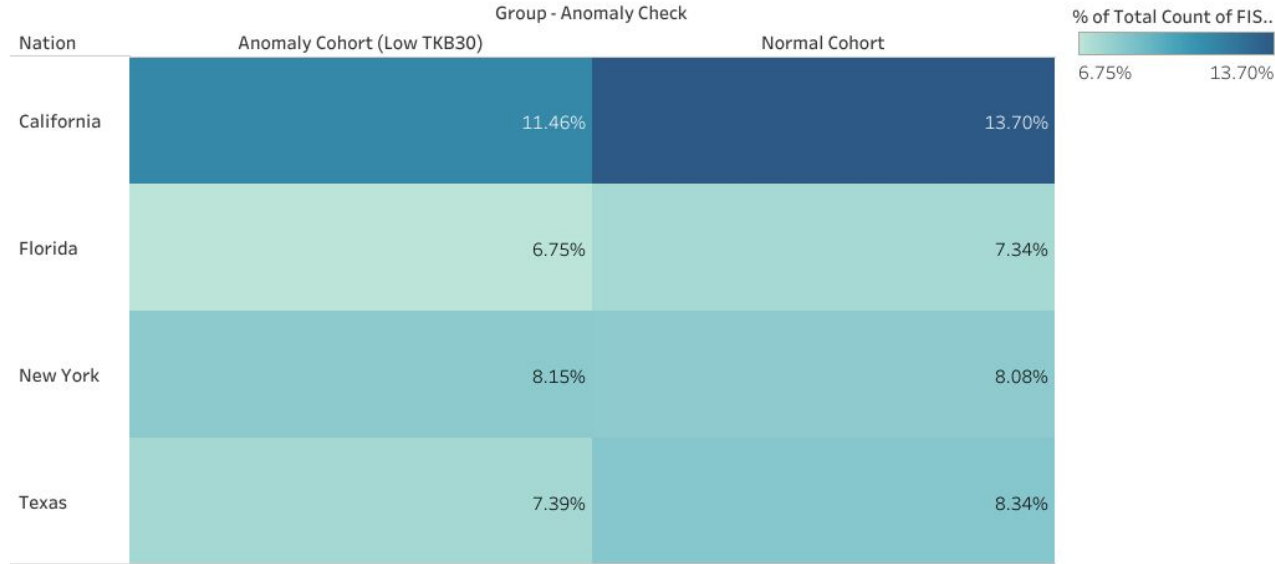
It turns out that many senior employees are also trapped in debt (over-leveraged), perhaps because they feel secure and therefore dare to take on a lot of debt.

The anomaly cohort appears heterogeneous

The presence of many borrowers with 10+ years alongside performance issues indicates that other factors override job stability

State Analyst

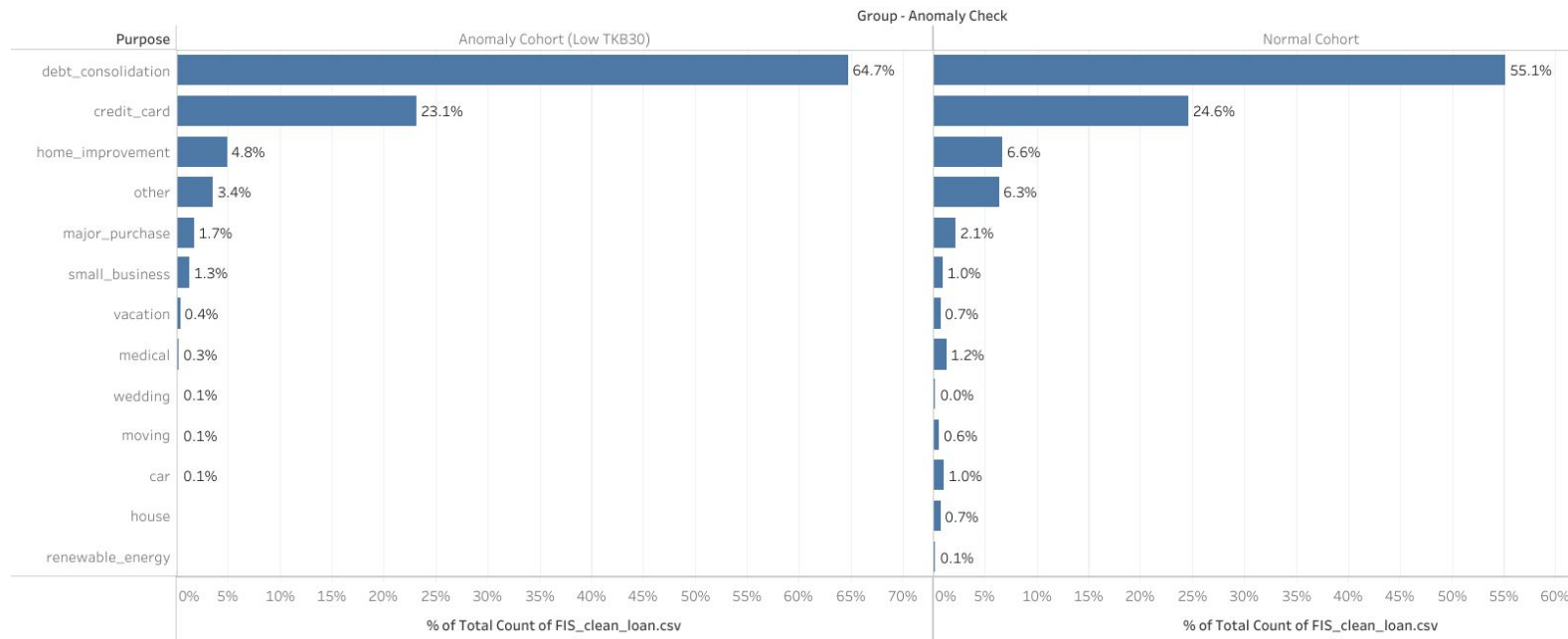
Anomaly vs Normal Customer



Interstate distribution is similar

No single state is clearly the source of the anomaly. The percentages for California, Florida, New York, and Texas differ only by a few points between the Anomaly (Low TKB30) and Normal Cohort. This means that the poor pattern is not concentrated in one large state it is **not** a single **geographical issue**.

Purpose Analyst

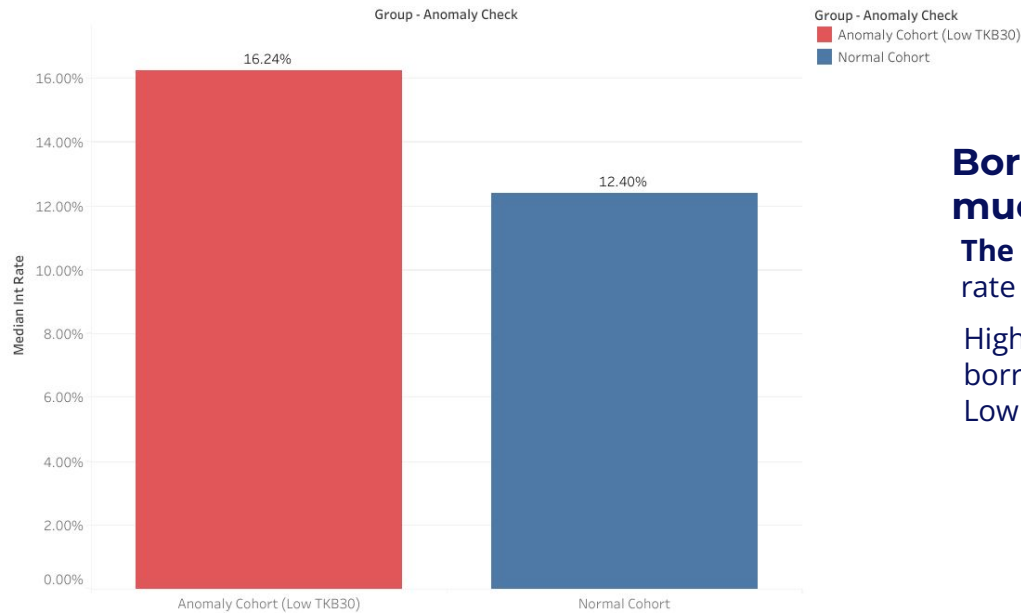


Debt Consolidation dominates the Anomaly Cohort

Borrowers who take out loans to **“tidy up”** their debts are more likely to **fall** into **the Low TKB30 category**.

Interest Rate Analyst

Anomaly vs Normal Customer



Borrowers with higher interest rates are much riskier.

The Anomaly Cohort has a **much higher** median interest rate (16.24%) than the Normal Cohort (12.40%).

High interest rates increase **installment burdens**, making borrowers more prone to **delinquency** and **falling** into the Low TKB30 category.



03 Insight & Recommendation

Summary Insight



The Pricing Paradox

We give the highest interest rates (~16%) to people with the thinnest wallets (lower salaries & mortgage payments).



The “Senior Employee” Trap

Many customers with bad credit have been working for 10+ years.

Mortgage is a Liability, Not an Asset

Customers with MORTGAGE status dominate bad credit.



The “Refinancing” Loop

The majority of non-performing loans are for debt consolidation purposes.



RECOMENDATIONS



Short-Term (Months 1-2) - **Stop the Bleeding**

Limit the Pricing Cap, Specifically for Debt Consolidation purposes



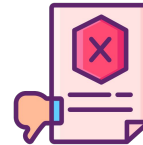
**Volume Lost
Opportunity**

\$566.767.975



**Bad Debt
Saved**

\$20.842.925



**Bad Rate
Rejected**

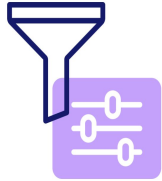
3.68%



**Benchmark
Normal Rate**

1.44%

RECOMENDATIONS



Medium-Term (Months 3-4) - **Fix the Filter**

You don't have to include the "Home Ownership Credit" factor as a deduction from your salary in order to know how much money you actually have left.

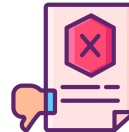
Threshold \$77673.0



**Volume Lost
Opportunity
\$615.055.050**



**Bad Debt
Saved
\$111.80.075**



**Bad Rate
Rejected
1.82%**



**Benchmark
Normal Rate
1.91%**

RECOMENDATIONS



Long-Term (Months 5-6) - **Pivot to Quality**

Target Segment Shift the marketing focus from Consolidation to Home Improvement/Credit Cards for homeowners who have paid off their debts.

Compare Bad Rate



**Debt
Consolidation**
2.05%



**Home
Improvement**
1.72%



**Risk
Reduction**
0.33%



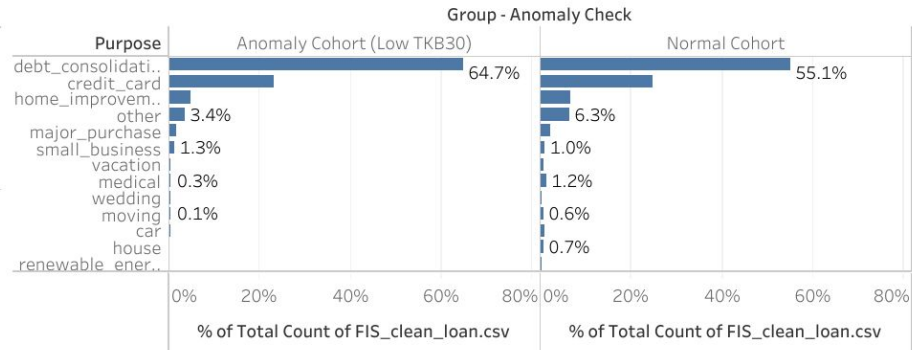
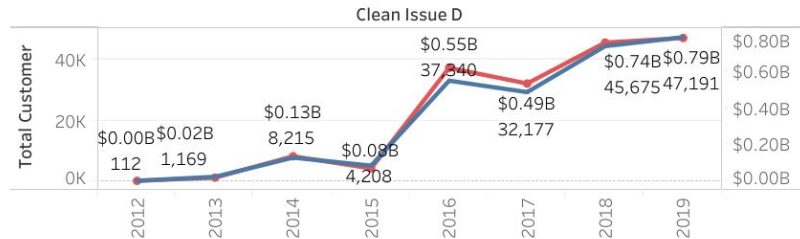
**Potential
Savings**
\$328.045.12

DASHBOARD



Portfolio Optimization: Revamping Risk-Based Pricing Strategy Addressing the 59% High-Risk Concentration in Debt Consolidation Segment (6-Month Roadmap)

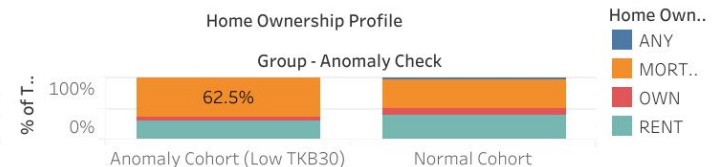
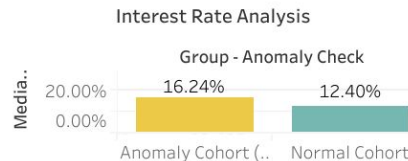
Total Customer	Outstanding	ENR(OS Bersih)	TKB30
176,087	\$2.80B	\$2.80B	98.11%



Measure Names
 ■ Outstanding
 ■ Total Customer



Group - Anomaly Check
 ■ Anomaly Cohort (Low..
 ■ Normal Cohort



Thank You



Contact me Mikhael Putra Iswanto



APPENDIX

Structured Query Language

01
Data
Cleaning

02
EDA &
Aggregate

03
Anomaly vs
Normal

04
DPD, KOL, AUM

05
Impact Analyst